INCH-POUND
MIL-P-70874 (AR)
10 March 1992

MILITARY SPECIFICATION

PRINTED WIRING BOARD ASSEMBLY, TOW TRIGGER INTERFACE SPECIFICATION FOR

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for manufacture, inspection, packaging and acceptance of the Printed Wiring Board Assembly (PWBA), TOW Trigger Interface, part number 9339658.

Beneficial comments (recommendations, addition, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1265

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-F-13926	-	Fire Control Materiel Manufacture and Inspection of, General Specification for
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-W-63150	-	Weapons and Support Materiel, Standard Quality Assurance Provisions for

STANDARDS

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-454	-	Standard General Requirements for Electronic Equipment
MIL-STD-810	-	Environmental Test Methods
MIL-STD-1686	-	Electrostatic Discharge Control Program for Protection of Electrical and Electronic Parts, Assemblies and Equipment (Excluding electrically initiated explosive devices)
MIL-STD-1695	-	Environments, Working, Minimum Standards for
MIL-STD-2000	-	Standard Requirements for Soldered Electrical and Electronic Assemblies
MIL-STD-45662	-	Calibration System's Requirements

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HANDBOOK

MILITARY

DoD-HBK-263 - Electrostatic Discharge Control Handbook for Protection of Electrical and Electronic Parts, Assemblies and Equipment (Excluding electrically initiated explosive devices) METRIC

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER.

PRODUCT DRAWINGS (CAGE CODE 19200)

9339658	-	Printed Wiring Board Assy, TOW Trigger Interface
9339660	-	Schematic Diagram, TOW Trigger Interface

PACKAGING DATA SHEET (CAGE CODE 19200)

SPI-P9339658	-	Printed Wiring Board Assy - TOW Trigger Interface

(Copies of other Government documents, drawings and publications required by the contractor in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Item description. This printed wiring board assembly contains the interface circuitry necessary to select a simulated TOW missile and to produce a trigger output pulse. The TOW wire cut signal is generated in response to an input signal. Rocket Trigger and WESS trigger pulses are also generated in response to an input command. Test and kill for the various outputs completes the functions implemented on this board.

Circuitry required to perform these functions includes logic elements, monostable multivibrators and discrete components.

3.2 First article. When specified (see 6.4), a sample shall be subjected to first article inspection in accordance with 4.4.

3.3 Environmental stress screen (ESS). All assemblies shall be subjected to an environmental stress screen prior to conformal coating consisting of temperature cycling and soaking between the temperatures -35°C and +62.8°C. After exposure to ESS in accordance with 4.5, the characteristics of 3.4 shall not be degraded.

3.4 Characteristics.

3.4.1 Operating conditions. The operating conditions shall be in accordance with Table I, measured under quiescent conditions. For terminal and component designations called out in this specification, see drawing 9339660, Schematic Diagram, TOW Trigger Interface.

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TABLE I. Operating Conditions.

Parameter	Terminal	Minimum	Nominal	Maximum	Unit
Ambient Temperature		-31.7	25	55	°C
Input (+)	J1-19	24	28	32	Vdc
Power (+)	J1-27	8	12	13	Vdc
Return (-)	J2-2				
Power (-)	J1-24				
Input	J1-19			12	mA
Current	J1-27			5	mA

3.4.2 Performance. Unless otherwise specified, the PWBA shall meet the requirements of this specification under the following conditions:

a.	Temperature	25°±10°C (77°±18°F)
b.	Relative humidity	50±30 %
c.	Barometric pressure	725+50, -75mm Hg (28.5+2.0, -3.0 inches Hg)
d.	Input power and loads	See Figure 1

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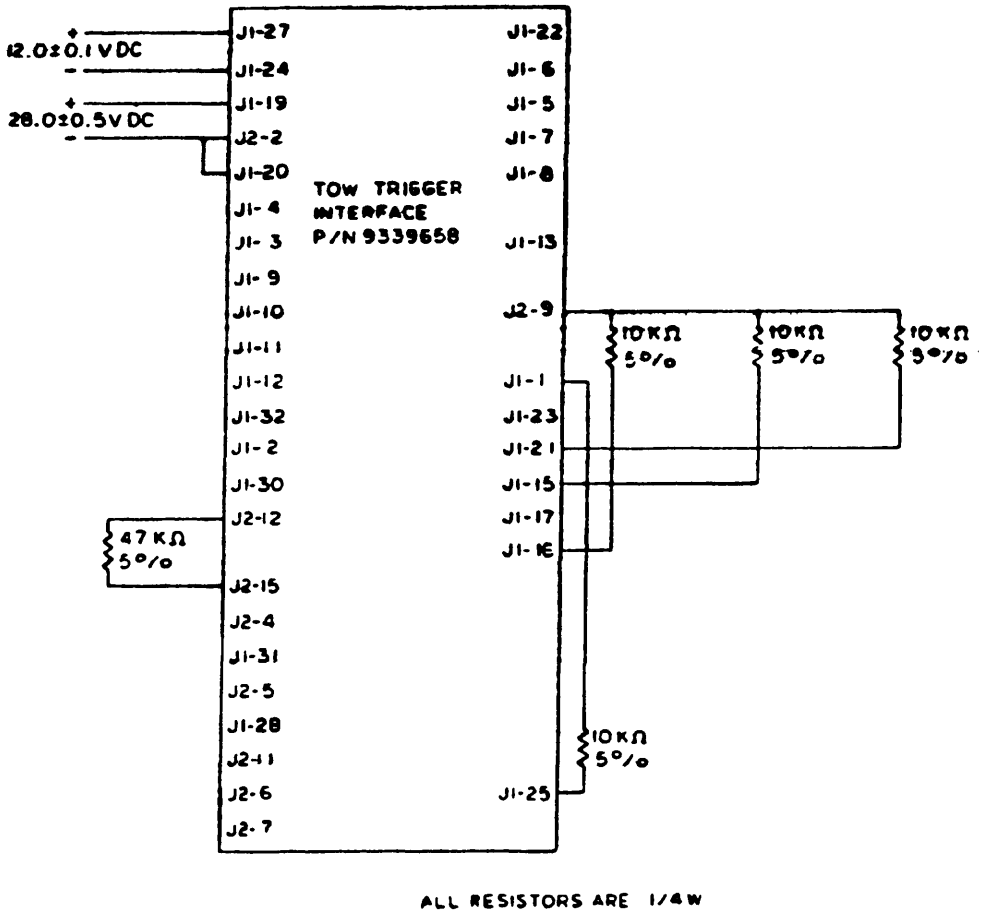


FIGURE 1. TOW Trigger Interface PWBA Power Source and Loads.

3.4.2.1 Current drain. The quiescent current drain shall be 12 mA maximum at terminal J1-19 and 5mA maximum at terminal J1-27.

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3.4.2.2 Select none. With power applied and missile, terminals J1-9, J1-10, J1-11, and J1-12 connected to terminal J1-19, four transistors in U2 turn on and send low voltage levels to U3, and U3 send high voltage levels to pins 8, 12, 5, and 1 of "AND" U4. The voltage at terminals J1-8, J1-7, J1-6 and J1-5 shall be 0.5 ± 0.5 Vdc. The voltage at terminal J1-22 shall be 12.0 ± 0.25 Vdc due to voltage regulator U7.

3.4.2.3 TOW fire. With power applied, and terminals J1-32 (TOW Fire) and J1-9 (Missile 1) connected to terminal J1-19, the voltage at terminals J1-5, J1-7, and J1-8 shall be 0.5 ± 0.5 Vdc, and at terminal J1-6 the voltage shall be 11.5 ± 0.6 Vdc. At terminal J1-23 (TOW Trigger) the voltage shall be a positive pulse with a 10.5 ± 1.0 V amplitude and 120 ± 40 ms width (after J1-32 is connected to terminal J1-19). The voltage at J1-6 is high since J1-9 pulls pin 8 of "AND" U4 high and relay K1 at pin 4 turns on with the TOW Fire, pulling pin 9 of "AND" U4 high. The TOW Fire also activates monostable U11 which is set by capacitor C21 and resistor R30, to provide a 120 ms pulse.

3.4.2.4 Trigger inhibit. With power applied, terminals J1-10 (Missile 2) and J1-32 (TOW Fire) connected to terminal J1-19, and terminal J1-31 (Trigger Inhibit) connected to terminal J2-15, the voltage at terminals J1-6, J1-7 and J1-8 shall be 0.5 ± 0.5 Vdc, at terminal J1-5 the voltage shall be 11.5 ± 0.6 Vdc and at terminal J1-23 (TOW Trigger) shall be a voltage of 0.3 ± 0.3 Vdc. The voltage at terminal J1-5 is high since J1-10 pulls pin 12 of "AND" U4 high and relay K1 at pin 4 turns on with the TOW Fire and pulls pin 13 of "AND" U4 high. The TOW Fire did not activate monostable U11, since the inhibit signal grounded the inputs of "NAND" U8 at pins 12 and 13 which would have triggered U11.

3.4.2.5 Trigger kill. With power applied, terminals J1-11 (Missile 3) and J1-32 (TOW Fire) connected to terminal J1-19, terminal J1-30 (Kill) connected to J1-2, the voltage at terminals J1-5, J1-6 and J1-8 shall be 0.5 ± 0.5 Vdc, the voltage at terminal J1-7 shall be 11.5 ± 0.6 Vdc and at terminal J1-23 the voltage shall be 0.3 ± 0.3 Vdc. The voltage at terminal J1-7 is high since J1-11 pulls pin 5 of "AND" U4 high and relay K1 at pin 4 turns on with the TOW Fire and pulls pin 6 of "AND" U4 high. The TOW Fire did not activate monostable U11, since the kill signal resets U11.

3.4.2.6 Select missile. With power applied, and terminals J1-12 (Missile 4) and J1-32 (TOW Fire) connected to terminal J1-19, the voltage at terminals J1-5, J1-6 and J1-7 shall be 0.5 ± 0.5 Vdc and at terminal J1-8 the voltage shall be 11.5 ± 0.6 Vdc.

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The voltage at terminal J1-8 is high since J1-12 pulls pin 1 of "AND" U4 high. Relay K1 at pin 4 turns on with the TOW Fire and pulls pin 2 of "AND" U4 high.

3.4.2.7 Wire cut. With power applied, and terminal J1-4 (Wire cut) connected to J1-19, at terminal J1-16 shall be a negative pulse from 10 ± 1 V, to 0.5 ± 0.5 V, to 10 ± 1 V and have a 145 ± 50 ms width (after the terminals are connected). The wire cut input turns on optical isolator U6 which triggers timer U10. The timer is set by resistors R28 and R27 and capacitor C20, to provide a 145 ms pulse.

3.4.2.8 Emergency wire cut. With power applied, and terminal J1-3 (Emrgency wire cut) connected to J1-19, at terminal J1-16 shall be a negative pulse from 10 ± 1 V, to 0.5 ± 0.5 V, to 10± 1 V and have a 145 ± 50 ms width (after the terminals are connected). The wire cut input turns on optical isolator U6 which triggers timer U10. The timer is set, by resistors R28 and R27 and capacitor C20, to provide a 145 ms pulse.

3.4.2.9 TOW trigger. With power applied, and terminal J2-5 (TOW trigger test) connected to J2-7, at terminal J1-23 (TOW trigger) shall be a positive pulse with a 10.5 ± 1.0 V amplitude and 120 ± 40 ms width. The TOW trigger activates monostable U11 which is set, by capacitor C21 and resistor R30, to provide a 120 ms pulse.

3.4.2.10 Rocket trigger. With power applied, and terminal J2-12 (Rocket trigger) connected to J1-2, at terminal J1-21 (Rocket transmitter trigger) shall be a negative pulse from 10 ± 1 V, to 0.5 ± 0.5 V, to 10 ± 1 V and have a 12 ± 4 ms width, and at terminal J1-1 (WESS trigger) shall be a positive pulse of 28.0 ± 0.5 V amplitude and 750 ± 250 ms width (after the terminals are connected). The rocket trigger activates monostable U11 at pin 4 which is set, by resistor R31 and capacitor C23, to provide a 12 ms pulse. The output at pin 7 of the monostable then triggers timer U10. The timer U10 triggers K2 to provide a pulse width of 750 ± 250 ms at J1-1. The timer U10 is set, by resistors R40 and R41 and capacitor C28, to provide a 750 ms pulse.

3.4.2.11 Rocket enable. With power applied, and terminal J2-11 (Rocket trigger test) connected to J2-7, at terminal J1-21 (Rocket transmitter trigger) shall be a negative pulse from 10 ± 1 V, to 0.5 ± 0.5 V, to 10 ± 1 V and have a 12 ± 4 ms width (after the terminals are connected), and at terminal J1-1 (WESS trigger) shall be a voltage of 0.0 (+0.01, -0.0) Vdc. The rocket trigger test signal activates monostable U11 at pin 5 which is

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set, by resistor R31 and capacitor C23, to provide a 12 ms pulse. The output at pin 7 of the monostable sends a trigger signal to pin 6 of timer U10, which is deactivated by the rocket trigger test signal.

3.4.2.12 WESS enable. With power applied, and terminal J2-4 (WESS test) connected to J2-7, at terminal J1-21 (Rocket transmitter trigger) shall be a voltage of 10.5 ± 1.0 Vdc and at terminal J1-1 (WESS trigger) shall be a positive pulse with a 28.0 ± 0.5 V amplitude and 750 ± 250 ms width (after the terminals are connected). The WESS test signal triggers timer U10 at pin 6 which is set, by resistors R40 and R41 and capacitor C28, to provide a 750 ms pulse. The output (pin 5) of timer U10 sends a pulse to pin 16 of U12. A transistor of U12 grounds pin 6 of K2 which provides a pulse width of 750 ± 250 ms at J1-1. The monostable U11 is not sent a trigger so the rocket transmitter trigger terminal is kept high.

3.4.2.13 Rocket disable. With power applied, and terminals J1-30 (Kill) and J2-12 (Rocket trigger) connected to J1-2, the voltage at terminal J1-21 (Rocket transmitter test) shall be 10.5 ± 1.0 Vdc and the voltage at terminal J1-1 (WESS trigger) shall be 0.0 (+0.01, -0.0) Vdc. The Rocket trigger signal would normally trigger monostable U11 at pin 4 which would then trigger timer U10 at pin 6. The kill signal, however, resets monostable U11 and disables timer U10, thus preventing an output pulse at J1-1 (WESS trigger).

3.4.2.14 Detector. With power applied, and terminal J1-28 (Detector signal) connected to J2-15, the voltage at terminal J1-17 (Signal buffered) shall be 1.5 ± 0.5 Vdc and the voltage at terminal J1-15 (Controller key) shall be 10.5 ± 1.0 Vdc. The detector signal turns on the transistor at pin 3 of U12 which grounds terminal J1-17. The voltage at terminal J1-15 is high since the transistor at pin 13 of U12 is off.

3.4.2.15 Detector disable. With power applied, and terminal J2-6 (Controller key) connected to terminal J2-7, the voltage at terminal J1-17 (Signal buffered) shall be 10.5 ± 1.0 Vdc and the voltage at terminal J1-15 (Controller key) shall be 3.0 ± 0.5 Vdc. The voltage at J1-17 is high since the transistor at pin 3 of U12 is off. The controller key signal turns on the transistor at pin 13 of U12. The voltage drops across the external 10 kohm resistor, connected to 11 V and terminal J1-15, resistor R44, and diodes CR14 and CR15, makes the voltage at J1-15 about 3V.

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3.4.3 Environmental requirements. Unless otherwise specified the assembly shall be subject to the following environmental conditions and meet the stated requirements.

3.4.3.1 High temperature. The assembly shall not be damaged and shall meet the performance requirements of 3.4.2. when exposed to and after the following.

3.4.3.1.1 Operating temperature. The assembly shall operate up to a temperature of +55°C (+130°F).

3.4.3.1.2 Storage temperature. The assembly shall operate after exposure to a temperature of +62.8°C (+145°F) for a period of 48 hours.

3.4.3.2 Low temperature. The assembly shall not be damaged and shall meet the performance requirements of 3.4.2 when exposed to and after the following.

3.4.3.2.1 Operating temperature. The assembly shall operate down to a temperature of -31.7°C (-25°F) after exposure at -31.7°C for 4 hours.

3.4.3.2.2 Storage temperature. The assembly shall operate after exposure to a temperature of -34.4°C (-30°F) for a period of 24 hours.

3.4.3.3 Shock. The assembly shall meet the performance requirements of 3.4.2 after shocks with a nominal level of 40 g's terminal peak and nominal duration of 11 milliseconds as shown in Figure 2. Apply three shocks (18 total) in each direction of the three mutually perpendicular axes.

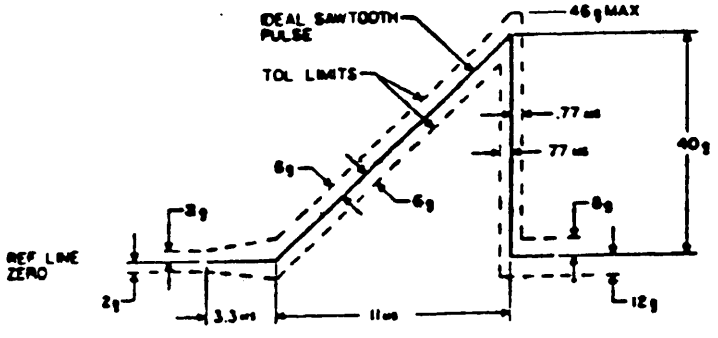


FIGURE 2. Shock Pulse.

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3.4.3.4 Vibration. The assembly shall meet the performance requirements of 3.4.2 after sinusoidal vibrations of frequency and amplitude given below and in Figure 3 on three mutually perpendicular axes. Total duration shall be 60 minutes per axis.

Frequency	Amplitude	Sweep rate
7-30 Hz	±1.5 g	7-500-7 Hz
30-50 Hz	0.033" DA	in 15 minutes
50-500 Hz	±4.2 g	

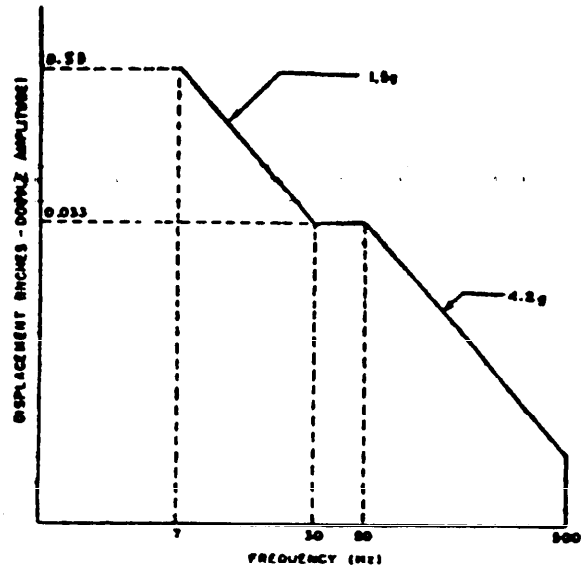


FIGURE 3. Vibration Test.

3.5 Design and construction. The PWBA shall be fabricated and assembled in accordance with drawings, parts lists, and other documents specified on drawing 9339658. The PWBA shall be manufactured and tested in a working environment in accordance with Table I, work area 5 of MIL-STD-1695.

3.5.1 Standards of manufacture. Handling of Electrostatic Sensitive Parts. This equipment contains semiconductor devices such as CMOS, mosfets, and microelectronic devices which may be degraded or damaged by an inadvertent discharge of static electricity. To prevent damage to these components, the supplier shall provide for adequate handling procedures and equipment in accordance with MIL-STD-1686 and DoD-HBK-263.

3.5.2 Soldering. Soldering shall be in accordance with the general requirements of section 4 of MIL-STD-2000.

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3.5.3 Workmanship. General workmanship shall be in accordance with requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Environmental stress screen (see 4.5).

c. Quality conformance inspection (see 4.6).

d. Packaging inspection (see 4.6.4).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the working environment specified in 3.5 and at standard ambient conditions defined in 3.4.2 using the methods of 4.8 and 4.9.

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4.4 First article inspection.

4.4.1 Sample. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.4.2. The first article sample shall consist of three printed wiring board assemblies.

4.4.2 Inspections to be performed. First article inspections are listed in Tables II and III. Each first sample unit shall be inspected in accordance with the Table II before being subjected to the environmental tests in Table III in the sequence listed. As determined by the Government, the first article assemblies may be subjected to any or all of the examinations and tests specified in this detail specification and be inspected for compliance with any or all requirements of the applicable drawings.

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TABLE II. Examinations and Performance Tests. 1/ 2/

Class	Characteristics	Require-ments Paragraph	Test/ inspec-tion Method	First Article Sample Size	Quality 3/ Conformance Inspection Sample Size/ Accept Criteria
101	Current Drain	3.4.2.1	4.8.1.1	3	32/Acc-0,Rej-1
102	Select None	3.4.2.2	4.8.1.2	3	32/Acc-0,Rej-1
103	TOW Fire	3.4.2.3	4.8.1.3	3	32/Acc-0,Rej-1
104	Trigger Inhibit	3.4.2.4	4.8.1.4	3	32/Acc-0,Rej-1
105	Kill Trigger	3.4.2.5	4.8.1.5	3	32/Acc-0,Rej-1
106	Select Missile	3.4.2.6	4.8.1.6	3	32/Acc-0,Rej-1
107	Wire Cut	3.4.2.7	4.8.1.7	3	32/Acc-0,Rej-1
108	Emergency Cut	3.4.2.8	4.8.1.8	3	32/Acc-0,Rej-1
109	TOW Trigger	3.4.2.9	4.8.1.9	3	32/Acc-0,Rej-1
110	Rocket Trigger	3.4.2.10	4.8.1.10	3	32/Acc-0,Rej-1
111	Rocket Enable	3.4.2.11	4.8.1.11	3	32/Acc-0,Rej-1
112	WESS Enable	3.4.2.12	4.8.1.12	3	32/Acc-0,Rej-1
113	Rocket Disable	3.4.2.13	4.8.1.13	3	32/Acc-0,Rej-1
114	Detector	3.4.2.14	4.8.1.14	3	32/Acc-0,Rej-1
115	Detector Disable	3.4.2.15	4.8.1.15	3	32/Acc-0,Rej-1
201	Workmanship	3.5.3	Visual	3	125/Acc-0,Rej-1
202	Soldering 4/	3.5.2	Visual	3	125/Acc-0,Rej-1
203	Conformal Coat	3.5	Visual	3	125/Acc-0,Rej-1
204	Dim Char.	3.5	SME 5/	3	125/Acc-0,Rej-1
205	Comp Orientation	3.5	Visual	3	125/Acc-0,Rej-1
206	Comp and Board Identification	3.5	Visual	3	125/Acc-0,Rej-1

NOTES:
1/ ESS testing per 4.5 is required for all deliverable units.
2/ 200 class (minor characteristics) examinations required prior to 100 class (major characteristics) performance tests.
3/ For lot sizes less than 32 or 125 as listed, 100% inspection is required (see 4.6.5).
4/ Check before conformal coating is applied.
5/ Standard Measuring Equipment.

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TABLE III. Environmental Tests.

Class	Characteristic	Requirements Paragraph	Test Paragraph	First Article Sample Size	Quality Conformance Inspection Sample Size/ Accept Criteria
116	High temp	3.4.3.1	4.9.1	3	1/Acc-0, Rej-1
117	Low temp	3.4.3.2	4.9.2	3	1/Acc-0, Rej-1
118	Shock	3.4.3.3	4.9.3	3	1/
119	Vibration	3.4.3.4	4.9.4	3	1/

NOTE:
1/ Not required for quality conformance inspection.

4.4.3 Rejection. If any assembly fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of any assembly to comply with any of the requirements.

4.5 Environmental stress screen. An environmental stress screening program shall be established. All units shall be subjected to a thermal stress screen in accordance with Table IV and the procedure below.

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TABLE IV. ESS Temperature Cycling Requirements.

Low Temp	High Temp	Rate Change	Power	Monitor	Cycles/ Group A Test
-34.4°C	+62.8°C	15°C/minute	Off	No	20/Failure Free or Additional Cycle

4.5.1 Conformal coating. Thermal stress screen shall be performed prior to conformal coating unless another method is approved by the Government.

4.5.2 Chamber temperature. Chamber temperature cycling shall be from -34.4°C (-30°F) to +62.8°C (+145°F) and the duration of one cycle to be 8 minutes up and 8 minutes down plus 4 minutes dwell at high temperature and 4 minutes at low temperature for a total cycle duration of 24 minutes. The units shall begin and end the cycles at ambient temperature (see Figure 4).

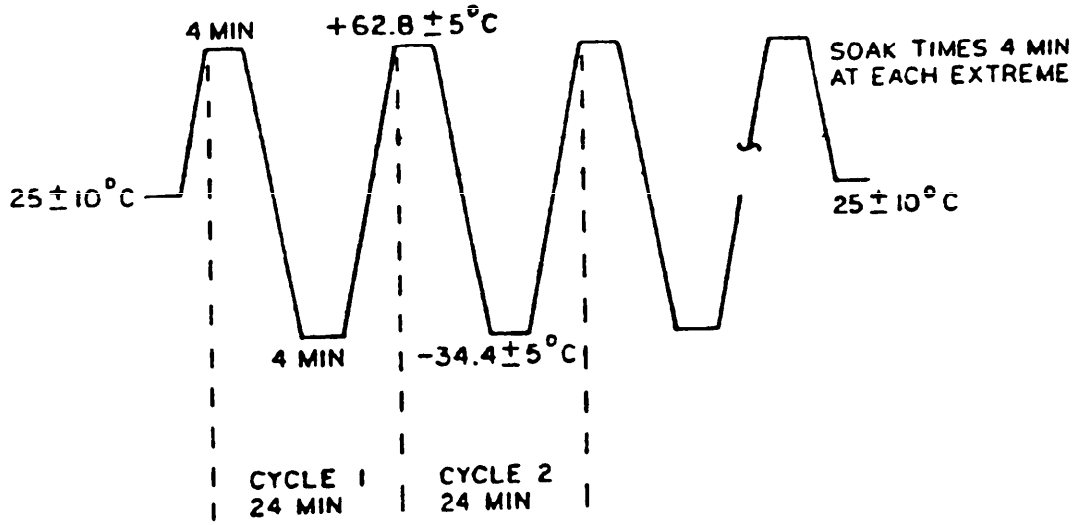


FIGURE 4. Environmental Stress Screen Temperature Profile.

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4.5.3 Cycle frequency. A total of twenty temperature cycles are to be imposed on each unit.

4.5.4 Temperature cycles. Temperature cycling of all units shall be conducted with units nonoperating and without power (static).

4.5.5 Test requirements. After the thermal stress screen, the units shall meet the Group A tests (4.6.2). If a unit is rejected, the unit shall be repaired for the failure (if possible) and be put through one additional cycle of ESS. The unit shall then undergo and pass the Group A tests.

4.6 Quality conformance inspection. Quality conformance inspection requirements are defined in 4.6.2 and 4.6.3.

4.6.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn of which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior of subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.6.2 Group A tests. Group A tests consist of examinations and performance tests listed in Table II in the quantities specified.

4.6.3 Group B tests. Group B tests shall consist of the tests indicated in Table III and shall be conducted on one sample unit per inspection lot which has been subjected to, and passed, Group A tests (see 4.6.2).

4.6.4 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall verify that the units are packaged in accordance with the requirements

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of 5.1. During the period of first article inspection, one additional unit, packaged in accordance with 5.1, shall be examined to the requirements for quality conformance inspection listed below. Failure of packaging shall be cause for rejection of packaging. Two units per lot shall be examined for the following:

a. Verify that the outside bag is marked in accordance with MIL-STD-129 and is properly closed.

b. Verify that sleeve is around box and that foam properly surrounds inner bag.

c. Verify that the inner bag is properly sealed.

d. Verify that the unit is not damaged after the removal of all packaging materials.

4.6.5 Failure of sample. Should any one sample fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract (see 6.7).

4.7 Inspection equipment.

4.7.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607 and MIL-W-63150.

4.7.2 Contractor-furnished inspection equipment.

4.7.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAP/QARs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs (see 6.3).

4.7.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the Methods of Inspection specified in 4.8 and 4.9 of this specification and with the component inspection procedures specified in Examination and Test Facilities requirements of MIL-F-13926, whenever Government designs are not specified (see 4.7.2.1). Since

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tolerance of test equipment is normally considered to be within ten percent of product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. When specified in contract or purchase order, the contractor shall submit inspection equipment designs for government review and evaluation (see 6.3).

4.7.2.3 Test equipment. In conjunction with 4.7.2.1 and 4.7.2.2, the test equipment listed in Table IV (Test equipment list) shall be used in the performance of the applicable test methods and procedures as specified in 4.8.

TABLE IV. Test Equipment List. 1/

Power Supplies (DC)	Standard - capable of providing levels and limits stated in 4.8.1.
DC Voltmeter/Multimeter	Standard - capable of measuring voltages stated in 4.8.1.
Ammeter/Multimeter	Standard - capable of measuring currents stated in 4.8.1.
Digital Oscilloscope	HP 54100 or equivalent - capable of measuring signals stated in 4.8.1.

NOTE:
1/ Must be calibrated to the requirements of MIL-STD-45662 prior to use.

4.8 Methods of inspection. Unless otherwise specified in the contract, the following procedures shall be utilized to conduct specified tests. If the contractor chooses to use alternate test procedures, a detailed description of the set-up and procedure shall be submitted to the Government procuring agency for approval prior to use (see 4.7.2.2).

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4.8.1 Performance tests. The following tests and examinations shall be performed to demonstrate compliance with 3.4.2. The PWBA shall be connected to the circuit specified in Figure 5. Apply power by turning ON S18 and S19 and leaving on until all tests have been completed. All other switches shall remain as shown in Figure 5 prior to each test specified in 4.8.1.1 through 4.8.1.15.

4.8.1.1 Current drain. Verify that the quiescent current drain meets the requirements of 3.4.2.1.

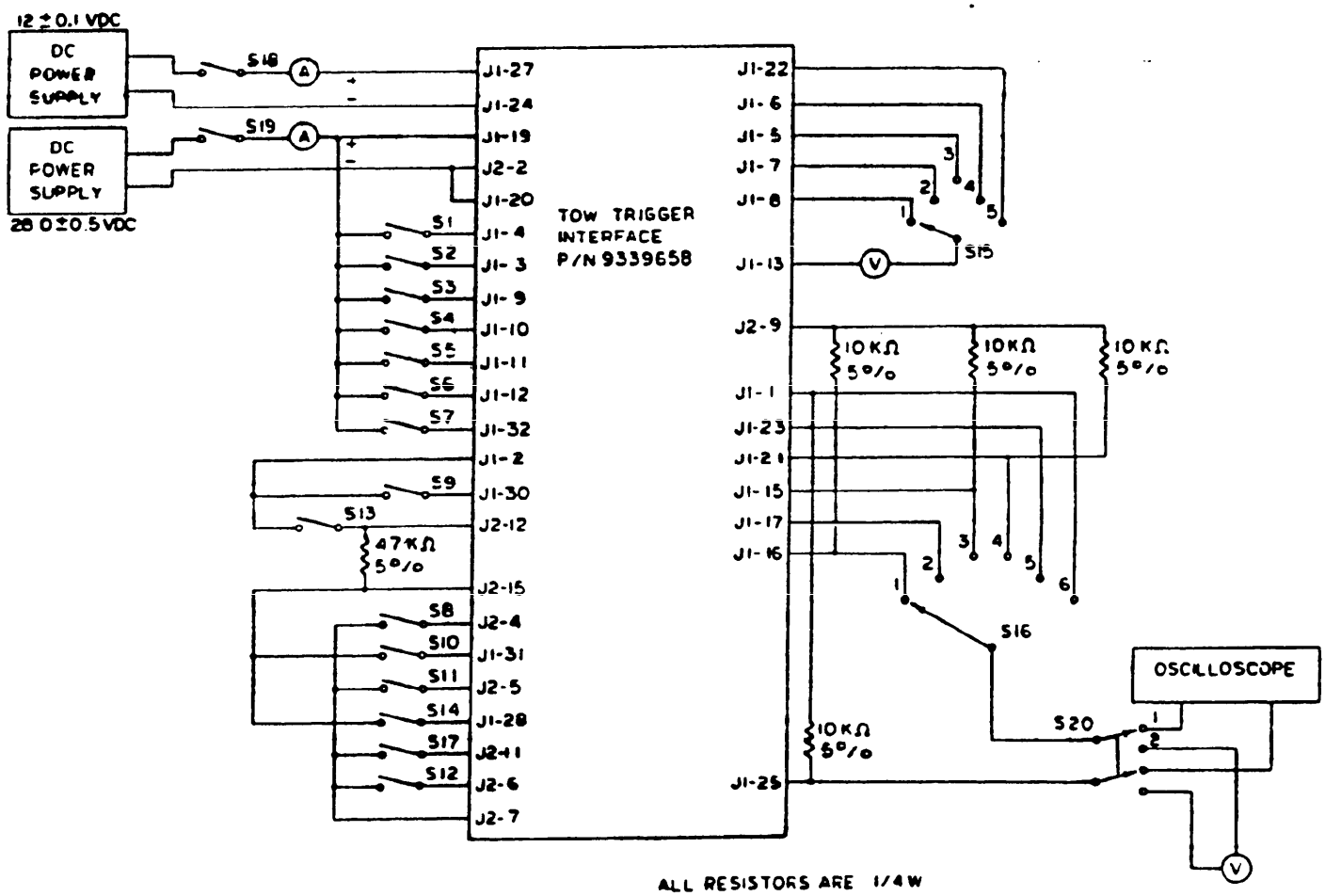


FIGURE 5. Test Circuit for TOW Trigger Interface.

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4.8.1.2 Select none. To verify the select none requirements of 3.4.2.2, turn ON S3, S4, S5 and S6 and measure the voltages at S15 in positions 1 through 5.

4.8.1.3 TOW fire. To verify the TOW fire requirements of 3.4.2.3, turn ON S7 and S3, set S20 to position 1 and measure the voltages at S15 in positions 1 through 4 and store the first 200 ms of the signal with the digital oscilloscope at S16 in position 5.

4.8.1.4 Trigger inhibit. To verify the trigger inhibit requirements of 3.4.2.4, turn ON S4, S7 and S10, set S20 to position 2 and measure the voltages at S15 in positions 1 through 4 and the voltages at S16 in position 5.

4.8.1.5 Trigger kill. To verify the trigger kill requirements of 3.4.2.5, turn ON S5, S7 and S9, set S20 to position 2 and measure the voltages at S15 in positions 1 through 4 and the voltage at S16 in position 5.

4.8.1.6 Select missile. To verify the select missile requirements of 3.4.2.6, turn ON S6 and S7 and measure the voltages at S15 in positions 1 through 4.

4.8.1.7 Wire cut. To verify the wire cut requirements of 3.4.2.7, turn ON S1, set S20 to position 1 and store the first 200 ms of the signal with the digital oscilloscope at S16 in position 1.

4.8.1.8 Emergency wire cut. To verify the emergency wire cut requirements of 3.4.2.8, turn ON S2, set S20 to position 1 and store the first 200 ms of the signal with the digital oscilloscope at S16 in position 1.

4.8.1.9 TOW trigger. To verify the TOW trigger test requirements of 3.4.2.9, turn ON S11, set S20 to position 1 and measure the signal at S16 in position 5.

4.8.1.10 Rocket trigger. To verify the rocket trigger requirements of 3.4.2.10, turn ON S13, set S20 to position 1 and store the first 20 ms of the signal with the digital oscilloscope at S16 in position 4. Repeat the test with S16 at position 6 and the oscilloscope set to store a second.

4.8.1.11 Rocket enable. To verify the rocket trigger requirements of 3.4.2.11, turn ON S17, set S20 to position 1 and store the first 20 ms of the signal with the digital oscillscope

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at S16 in position 4. Set S20 to position 2 and read the voltage at S16 in position 6.

4.8.1.12 WESS enable. To verify the rocket enable requirements of 3.4.2.12, turn ON S8, set S20 to position 1 and store the first second of the signal with the digital oscilloscope at S16 in position 6. Set S20 to position 2 and read the voltage at S16 in position 4.

4.8.1.13 Rocket disable. To verify the rocket disable requirements of 3.4.2.13, turn ON S9 and S13, set S20 to position 2 and measure the voltages at S16 in positions 4 and 6.

4.8.1.14 Detector. To verify the detector requirements of 3.4.2.14, set S20 to position 2, turn ON S14 and measure the voltages at S16 in positions 2 and 3.

4.8.1.15 Detector disable. To verify the detector disable requirements of 3.4.2.15, turn ON S12, set S20 to position 2 and measure the voltages at S16 in positions 2 and 3.

4.9 Environmental tests.

a. Unless otherwise specified, environmental tests shall be conducted in accordance with the test conditions and general test performance guidance of MIL-STD-810. Test facilities, chambers and apparatus used in conducting the test required by this specification shall be capable of meeting the conditions specified.

b. Where necessary in the following tests, it shall be permissible to temporarily open chamber doors to permit access to and manipulation of test equipment and sample.

4.9.1 High temperature. This satisfies the storage and operating conditions of high temperature requirements as specified in 3.4.3.

4.9.1.1 Storage and operating. The assembly shall be tested in accordance with MIL-STD-810, Method 501 using the following procedure:

a. Step 1 - Place the test item, configured for high temperature test, into the temperature chamber.

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b. Step 2 - Raise the internal temperature to 62.8°C (145°F).

c. Step 3 - Maintain the internal chamber temperature for a period of 48 hours while insuring the relative humidity is not in excess of 15 percent.

d. Step 4 - Adjust the internal chamber temperature to 55°C (130°F), the highest operating temperature for which the test item is designed to operate, and maintain until temperature stabilization of the item is reached.

e. Step 5 - Operate the test item until the item is stabilized and perform the tests required in 4.8.1.

f. Step 6 - Return the test item, nonoperating, to standard ambient conditions and stabilize.

g. Step 7 - Visually inspect for physical damage and in the absence of physical damage, perform the tests required in 4.8.1.

4.9.2 Low temperature. This satisfies the storage and operating conditions of low temperature requirements as specified in 3.4.3.

4.9.2.1 Storage and operating. The assembly shall be tested in accordance with MIL-STD-810, Method 502 using the following procedure:

a. Step 1 - Place the test item, configured for low temperature test, into the temperature chamber.

b. Step 2 - Lower the internal chamber temperature to -34.4°C (-30°F) and maintain for a period of 24 hours after stabilization.

c. Step 3 - Adjust the internal chamber temperature to -31.7°C (-25°F) and maintain the temperature for a period of 4 hours.

d. Step 4 - Operate the test item until the item is stabilized and perform the tests required in 4.8.1.

e. Step 5 - Return the test item, nonoperating, to standard ambient conditions and stabilize.

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f. Step 6 - Visually inspect for physical damage, and in the absence of physical damage, perform the tests required in 4.8.1.

4.9.3 Shock. The assembly, when mounted on a fixture that simulates its next higher assembly (see Figure 6), shall be tested in accordance with MIL-STD-810, Method 516.3, Section II, Procedure I, with the following modifications:

a. Step 3 - the assembly shall be in a non-operational mode.

b. The step 3 test transient shall be in accordance with 3.4.3.3.

c. Step 5 - the assembly inspection and functional verification shall not be conducted until shock has been completed in all three axes.

d. After shock testing the assembly shall not be damaged and shall be subjected to the performance tests specified in 4.8.1.

4.9.4 Vibration. The assembly, when mounted on a fixture that simulates its next higher assembly (see Figure 6), shall be tested in accordance with MIL-STD-810, Method 514.3, Section II, Procedure I, for a duration of 60 minutes with the following modifications:

a. Step 4 - mount the test item on vibration equipment.

b. Step 5 - expose the assembly to test level and durations as specified in 3.4.3.4.

c. Step 6 - does not apply.

d. Step 7 - the assembly inspection and functional verification shall not be conducted until vibration has been completed in all three axes.

e. After vibration testing the assembly shall not be damaged and shall be subjected to the performance tests specified in 4.8.1.

4.10 Certifications. Certification shall be provided that all units have been produced in accordance with the requirements

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of 3.5 and packaged in accordance with the requirements of 5.1.

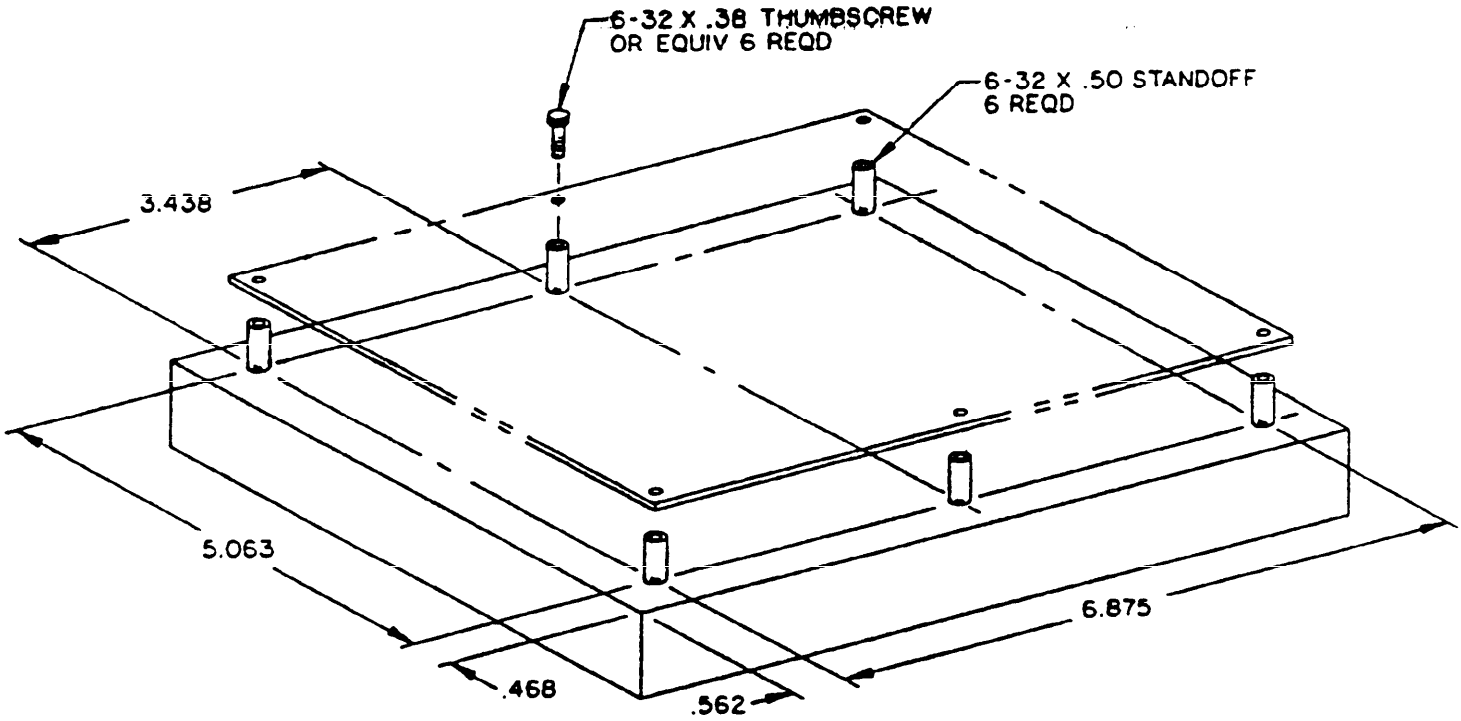


FIGURE 6. Mounting Configuration.

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5.0 PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with SPI-P9339658. The level of protection shall be A, B, or C, as defined by the procurement document.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The TOW Trigger Interface PWBA is part of the Multiple Integrated Laser Engagement System.

6.2 Acquisition requirements. Acquisition documents shall specify the following:

a. Title, number and date of this specification (see page 1).

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

c. Special tests and examinations required (see 4.2); the type of inspection required; and the activity responsible for testing, and its facility (see 4.1).

d. Whether a first article sample is required (see 3.2), and if so, pertinent details (see 4.4) including disposition of the first article sample.

e. Certification of conformance for each lot or shipment of product.

f. Whether Government-Loaned Property (Government Furnished inspection/test equipment) is available or will the contractor be required to furnish inspection equipment (see 4.7).

g. Applicable level(s) of preservation, packaging, packing and marking instructions (see 5.1).

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6.3 Submission of contractor inspection equipment designs for approval. Unless otherwise specified, submit two copies of designs as required to: Commander, U.S. Army AMCCOM, ATTN: SMCAR-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirement List, DD Form 1423, in the contract. The letter of submittal shall specify, the contractor; contract number; title, number and revision of this specification; item nomenclature; classification of defect and test paragraph.

6.4 Definitions.

6.4.1 Hit. Successful transmission of the weapon simulator kill code that results in a "hit" on the target, but that does not "kill" the target. An audio alarm at the target signifies the hit.

6.4.2 Kill. Successful transmission of the weapon simulator kill code that results in the deactivation of or "kill" of the target. An audio alarm at the target signifies the kill.

6.4.3 MILES. Multiple Integrated Laser Engagement System.

6.4.4 SME. Standard Measuring Equipment.

6.5 Drawings. Drawings listed in Section 2 of this specification under heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.6 Noncompliance. If a sample fails to pass group B inspection, the manufacturer shall notify the qualifying activity and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured with essentially the same materials and processes, and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the qualifying activity has been taken. After the corrective action has been taken, group B inspection shall be repeated on additional sample units (all tests and examinations, or the test which the original sample failed, at the option of the qualifying activity). Group

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tested in accordance with MIL-STD-810, Method 512.3, Proce

A inspections may be reinstituted; however, final acceptance and shipment shall be withheld until the group B inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity and the qualifying activity.

6.7 Subject term (key word) listing.

Fire Control
Multiple Integrated Laser Engagement System
Training Devices

Custodians:
Army - AR

Preparing activity:
Army - AR

(Project 1265-A032)

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-P-70874 (AR)	10 MARCH 1992

3. DOCUMENT TITLE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
US ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial (201) 724-6674	(2) AUTOVON 880-6674
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL,NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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